Advanced Programming Practice · 21CSC203P

Unit 5 · Session 6 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 6 : SLO â€“ 1 | Symbolic Calculus â€” Differentiation and Integration

I. Questions & Answers

1. Differentiate xÂ³ â€” d/dx(xÂ³) = 3xÂ². In SymPy: diff(x**3, x) â†’ 3*x**2.
2. Integrate 2x â€” âˆ« 2x dx = xÂ² + C. In SymPy: integrate(2*x, x) â†’ x**2 (constant omitted).
3. Differentiate sin(x) â€” d/dx sin(x) = cos(x). In SymPy: diff(sin(x), x) â†’ cos(x).

II. Code

- from sympy import symbols, diff, integrate, sin
- x = symbols('x')
- print(diff(x**3, x)) # 3*x**2
- print(integrate(2*x, x)) # x**2
- print(diff(sin(x), x)) # cos(x)